UNIT 9

INFORMATION HANDLING

AVERAGE

Define an average (arithmetic mean)

Consider the following example.

Example. Once Shahid Afridi made 6 runs in the first over, 10 runs in the second over, 8 runs in the third over, and 4 runs in the fourth over.

Now answer the following questions:

1. What was the number of his total runs?
$6 + 10 + 8 + 4 =$ **28 runs**

2. How many overs did he play?
4 overs

3. What was his run rate (runs per over)?
To answer this question, we find average as under:

$$\text{Run rate} = \frac{\text{Total runs made in all the overs}}{\text{Total numbers of overs played}}$$

$$= \frac{\overset{7}{\cancel{28}}\ \text{runs}}{\underset{1}{\cancel{4}}\ \text{overs}} = 7 \text{ runs per over}$$

The run rate 7 is the average score. This 7 represents his performance in all the overs.

Or

Average is the overall representative value of the information.

Average = Sum of the quantities divided by the total number of quantities

Find an average of given numbers.

Example. Find the average of 5, 8, 10, 12, and 20.

Solution:

Sum of given numbers = 5 + 8 + 10 + 12 + 20

$$\text{Average} = \frac{\text{Sum of quantities}}{\text{Total number of quantities}}$$

$$\text{Average} = \frac{5 + 8 + 10 + 12 + 20}{5} = \frac{55}{5}$$

$$= \frac{\cancel{55}^{11}}{\cancel{5}_{1}} = \frac{11}{1} = 11$$

Thus the average of the given numbers is **11.**

Activity 1

Find the average of first five even numbers.

Solution:

First five even numbers are: 2, 4, 6, 8, and 10.

$$\text{Average} = \frac{\text{Sum of numbers}}{\text{Total number}}$$

Solve it,

$$= \frac{2 + 4 + 6 + 8 + 10}{5} = \boxed{\frac{\quad}{\quad}} = \boxed{\quad}$$

Activity 2

Find the average of first five odd numbers.

Solution:

First five odd numbers are: 1, 3, 5, 7, and 9.

$$\text{Average} = \frac{\text{Sum of numbers}}{\text{Total number}}$$

Solve it,

$= \underline{\hspace{6cm}} = \boxed{\frac{\quad}{\quad}} = \boxed{\quad}$

EXERCISE 9.1

Q1. Find the average (mean) of the following numbers.

10, 12, 14, 16 and 18

Solution:

5, 8, 7, 9, 6, 4, 10

Solution:

2.0, 3.5, 4.0, 5.5, 6.5

Solution:

2, 3, 5, 7, 11, 13

Solution:

85, 90, 78, 88, 92, 95

Solution:

$\frac{7}{4}, \frac{9}{5}, \frac{11}{6}, \frac{13}{8}, \frac{15}{10}$

Solution:

$1\frac{2}{3}, 4\frac{5}{6}, 7\frac{8}{9}, 10\frac{11}{12}$ and $13\frac{14}{15}$

Solution:

Solve real life problems involving average.

Example 1. The attendance of class V during six days of a week was: 44, 40, 37, 42, 35, and 36. Find the average attendance?

Solution:

Daily attendance = 44, 40, 37, 42, 35, 36

$$\text{Average} = \frac{\text{Total attendance}}{\text{Number of days}}$$

$$\text{Average attendance} = \frac{44 + 40 + 37 + 42 + 35 + 36}{6} = \frac{234}{6}$$

$$= \frac{\cancelto{39}{234}}{\cancelto{1}{6}} = \frac{39}{1} = 39$$

Thus average attendance is **39.**

Example 2. The business train covers a distance of 450 km in 6 hours. What is its average speed?

Solution:

$$\text{Average speed} = \frac{\text{Distance covered}}{\text{Number of hours}} = \frac{450 \text{ km}}{6 \text{ hours}}$$

$$= \frac{\overset{75}{\cancel{450}} \text{ km}}{\underset{1}{\cancel{6}} \text{ hours}} = \frac{75 \text{ km}}{1 \text{ hr}} = 75 \text{ km per hr}$$

Thus average speed of the train will be **75 km/hr.**

EXERCISE 9.2

Q1. The maximum temperatures in Karachi recorded during the week were 32°C, 34°C, 33°C, 31°C, and 35°C. What was the average temperature during that week?

Solution:

Q2. A student tracks the number of steps taken each day for a week: 5,000 steps, 6,500 steps, 4,800 steps, 7,200 steps, 5,500 steps, 6,000 steps, and 5,700 steps. What is the average number of steps taken per day?

Solution:

Q3. In five days, the rainfall (in mm) in Lahore was recorded as 50 mm, 60 mm, 70 mm, 80 mm, and 90 mm. What is the average rainfall for the five days?

Solution:

Q4. A child receives a weekly allowance of Rs 50, Rs 60, Rs 70, Rs 80, and Rs 90. What is the average allowance received per week?

Solution:

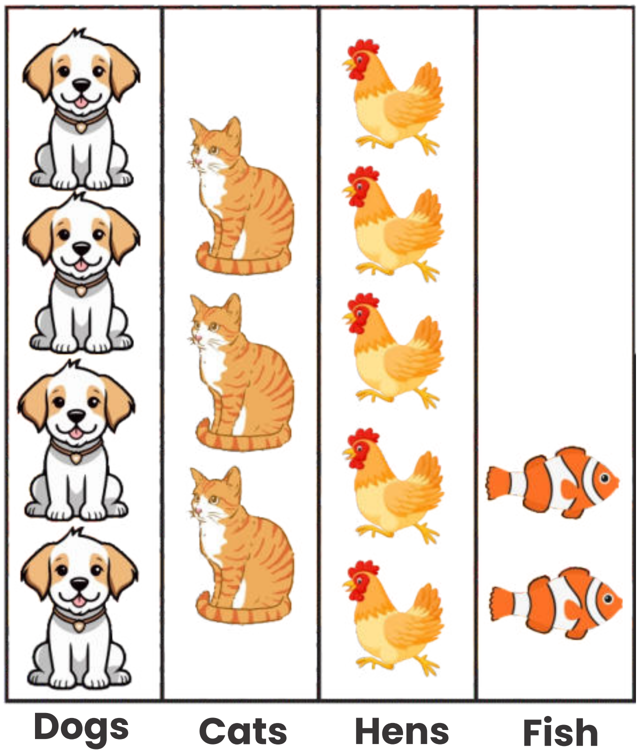
Q5. In a garden, the heights (in cm) of five plants are 20 cm, 25 cm, 30 cm, 35 cm, and 40 cm. What is the average height of the plants?

Solution:

BLOCK, COLUMN AND BAR GRAPH

Draw block graphs or column graphs

We have learnt picture graph in previous classes. Let us revise an example. The number of animals in Ali's farm house has been represented in picture graph

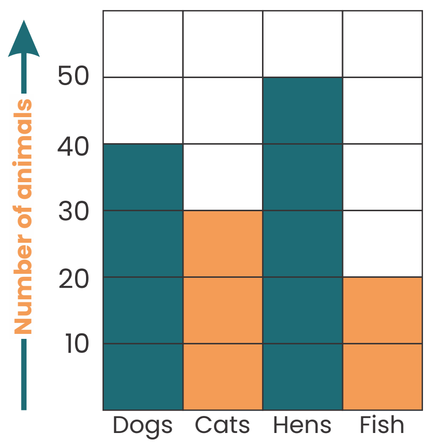


Each  represents **10 dogs.** Each  represents **10 hens.**

Each  represents **10 cats.** Each  represents **10 fish.**

This diagram is called **picture graph.** Picture graph helps us to see quantity of each thing/item at a glance and helps us to compare their differences. We can show these numbers of animals in blocks as:

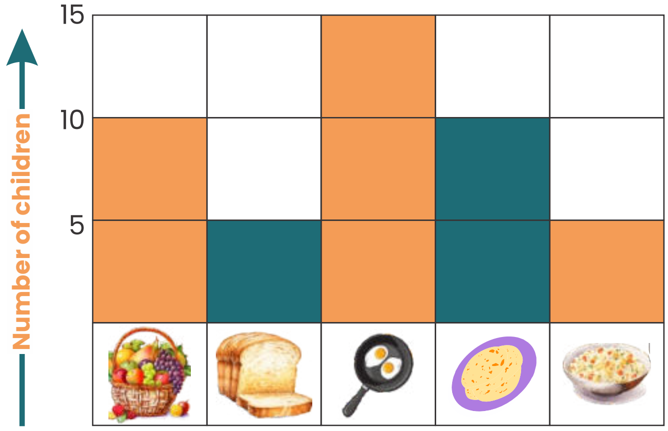
We call this a **block graph** or **column graph.** Column graph is used to represent small objects and quantities.



Example 1. Following table shows the result of the favourite breakfast food taken by class V students. Draw a block graph.

Fresh Fruit	Toast/Bread	Eggs	Paratha	Cereal
10	5	15	10	5

Solution:



Read the above graph and answer the following questions:

(i) Which breakfast food is most favourite?

(ii) Which breakfast food is equally favourite?

(iii) Which breakfast food is least favourite?

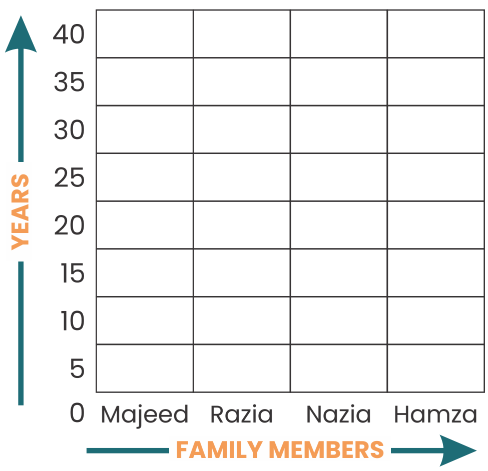
Activity

The ages of a family's members are as under. Help Sameer to draw the block graph.

- Majeed is 30 years old.
- Razia is 25 years old.
- Nazia is 10 years old.
- Hamza is 5 years old.

Step 1: Select the ages on vertical side and family members on horizontal side of the graph.

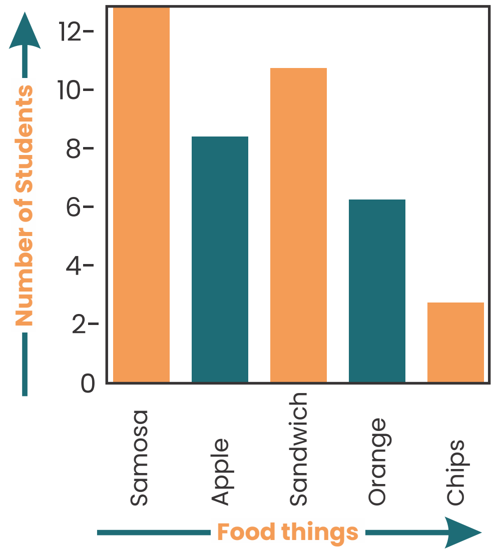
Step 2: Colour the columns to show the information.



Activity

Read the vertical bar graph and write the correct answer in the box.

This bar graph represents information about students as to what they ate at play time in school. The different food things are shown on horizontal line and number of students are shown on vertical line.



Answer the following questions:

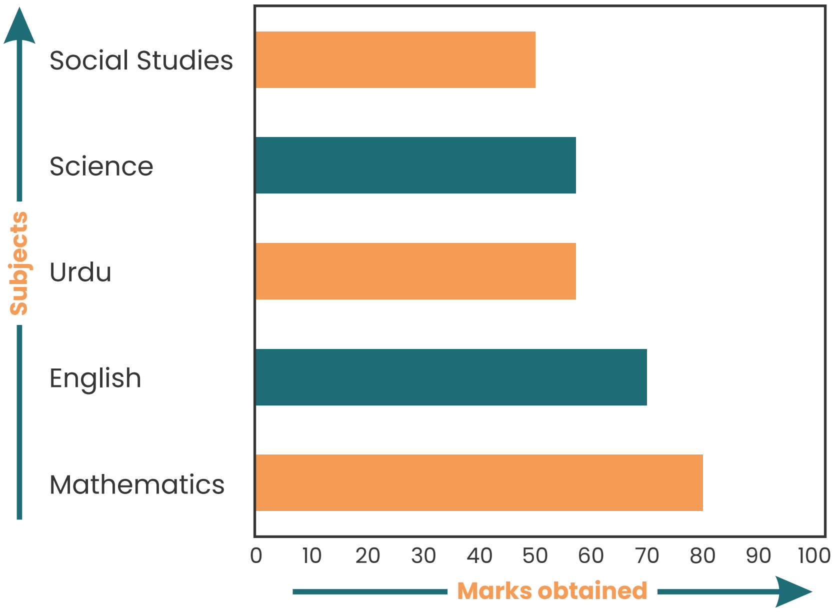
1. How many children ate samosa?	12
2. How many children ate oranges?	
3. How many children ate apples?	
4. How many children ate sandwiches?	
5. How many children ate chips?	

Activity

Read the following horizontal bar graph. In this graph, the marks obtained by Yasir of Grade V in five subjects are given.

Time in years is shown on horizontal axis. The number in millions is shown on vertical axis.

in five subjects are given



Read the horizontal bar graph and answer the following questions:

1. How many marks did Yasir get in English?	70
2. How many marks did he get in Urdu?	
3. How many marks did he get in Mathematics?	
4. How many marks did he get in Science?	
5. How many marks did he get in Social studies?	
6. In which subjects did he get the least number of marks?	
7. How many total marks did he get?	
8. In which subject did he get the highest marks?	
9. Which subjects did he get equal marks?	

EXERCISE 9.3

Q1. Draw a block graph or column graph of the following:

Number of books sold by a shopkeeper

Days	Sunday	Monday	Tuesday	Wednes	Thursday	Friday
Number of books sold	65	40	30	50	20	70

Solution:

Result of examination of class 6

Grade	A1	A	B	C	D
No. of Children	20	25	15	20	5

Solution:

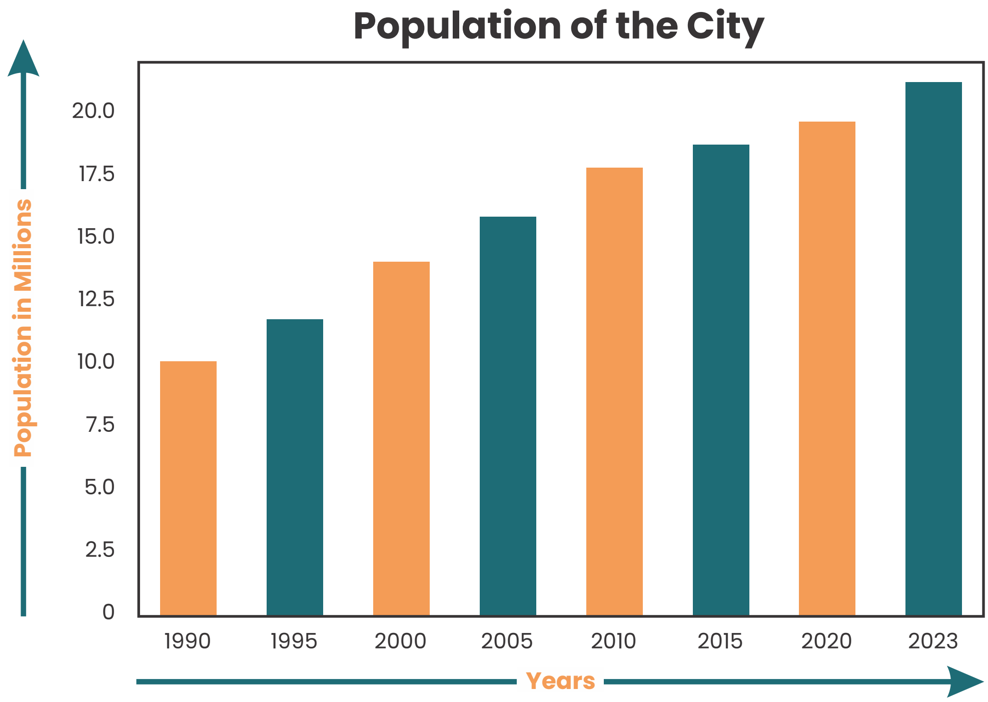
Marks obtained by amna during annual examination

SUBJECT	ISL	ENG	URDU	MATH	S.STD	SCI
Marks Obtained	80	50	65	90	40	55

Solution:

Q2. The population of Karachi city is represented in the following bar graph.

Years are shown on horizontal axis.
Population in millions is shown on vertical axis.



Read the graph and answer the following questions:

1. What was the population of Karachi in 1995?

2. What was the population of Karachi in 2000?

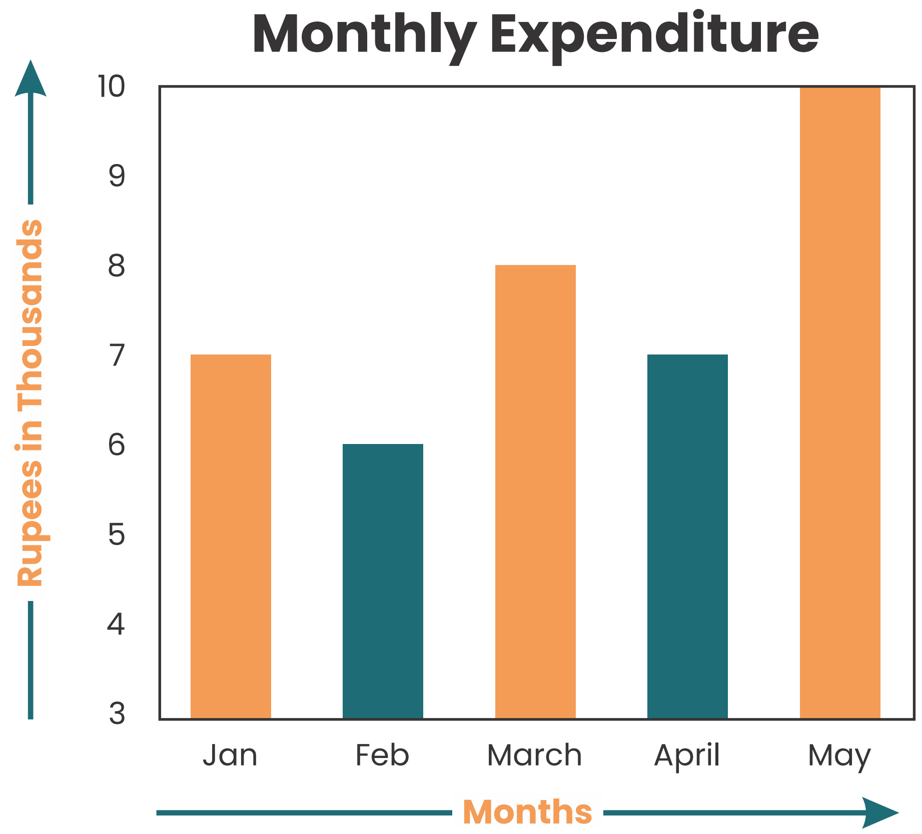
3. What was the population of Karachi in 2015?

4. In which year was the population least?

5. In which year was the population highest?

Q3. The expenditure of a family for five months is reported by the following bar graph.

Names of months are shown on horizontal axis.
Amount in thousand rupees is shown on vertical axis.



Look at the graph and answer the following questions:

1. In which month expenditure was the least?

2. In which month the expenditure was the greatest?

3. What was the expenditure in the month of February?

4. What was the expenditure in the month of April?

5. In which months the expenditure is equal?

6. In which month the expenditure is Rs 8000?

7. What is the amount of highest expenditure?

8. What is the amount of lowest expenditure?

DATA

Define and organize data.

(a) Definition of Data

The information collected from any field of study is called data. Mostly it is represented in the form of numerical figures.
Data can be obtained from existing sources or the same can be obtained directly from the target according to needs.

Example: Favourite colour of class IV students, different age groups in class V, marks obtained by someone in an examination, any office record, game records; etc.

(b) Organizing Data

There are many ways of collecting and organizing data.
Data can be represented through pictures, tables, charts, and graphs.

Example: The organizer of a market survey wants to know the ages of the customers.

Solution:

The ages of customers are:

17, 12, 15, 20, 25, 17, 19, 11, 12, 15, 20, 20, 12, 20, 22, 20, 19, 15

This way of representing the information does not help the organizer to answer any question.

Now we write the above information in ascending order:

11, 12, 12, 12, 15, 15, 15, 17, 17, 19, 19, 20, 20, 20, 20, 20, 22, 25

Though it is one way but it is time taking. There is possible that some data may be missed. However, it is easier to read data.
Now we organize the data in another way through tally method.

Age of customers in years	Tally marks	Number of customers
11	|	1
12	|||	3
15	|||	3
17	||	2
19	||	2
20	~~||||~~	5
22	|	1
25	|	1

Table (ii)

The organizer now reads the ages one by one from Table (I) and draws one small line called tally in the tally column against the age read in table (II). If a number occurs five times, he crosses the four tallies already drawn by a slash like this (卌). Thus (卌) stands for five numbers. Then counts the tallies and writes the numbers in the last column shown in the table.

Example:

The mathematics marks of class V students are as under:

Organize the data

32, 30, 31, 33, 33, 34, 34, 33, 32, 32, 34, 35, 35, 32, 33, 32, 31, 33, 32

Solution:

Now we write the above information in ascending order:

30, 31, 31, 32, 32, 32, 32, 32, 33, 33, 33, 33, 33, 34, 34, 34, 35, 35

The data is represented through tally chart.

Mark obtained	Tally marks	Number of students
30	|	1
31	||	2
32	卌 |	6
33	卌	5
34	|||	3
35	||	2

Activity

Organize the following information through tally method. The ages of students of class V in a school are given in the following table:

Ages of student in years										
	11	12	11	13	14	10	11	13	12	13
	12	13	11	11	14	10	12	12	11	13

Solution:

Writing the above information in ascending order

Ages of student in years										
	10		11						12	
				13					14	

Organize the given data through tally method.

Ages of students in years	Tally marks	No. of same ages of students
10		
11		
14		

EXERCISE

Q1. Organize the following survey information through tally method.

a. The favourite ice cream flavours of students in Class V are: Vanilla, Chocolate, Strawberry, Vanilla, Chocolate, Mango, Strawberry, Chocolate, Vanilla, Mango, Vanilla, Strawberry, Chocolate, Vanilla, Mango.

Solution:

b. The number of books read by students in a month are: 3, 5, 2, 4, 5, 3, 2, 4, 5, 3, 2, 4, 5, 3, 4.

Solution:

c. The daily screen time (in hours) of students is recorded as follows: 1, 2, 3, 1, 1, 2, 3, 2, 1, 2, 3, 1, 1, 3, 2.

Solution:

d. The number of pets owned by students in Class V are: 1, 2, 0, 1, 3, 0, 2, 1, 2, 3, 0, 1, 2, 1, 0.

Solution:

e. The number of sports played by students in a month are: 2, 3, 1, 2, 3, 2, 1, 3, 1, 2, 2, 3, 1, 2, 3.

Solution:

f. The number of homework assignments completed by students in a week are: 5, 4, 5, 6, 4, 5, 4, 6, 5, 4, 5, 4, 6, 5, 5.

Solution: